

Arhitektuur

See dokument sisaldab süsteemi arhitektuuri jooniseid MermaidJS formaadis (soovitav renderdada GitHubis või Mermaid Live Editoris).

Infrastruktuuri diagramm

```
graph LR
    %% Styles
    classDef user fill:#FFE4C4,stroke:#333,stroke-width:1px;
    classDef network fill:#E0F7FA,stroke:#00796B,stroke-width:1px;
    classDef cloud fill:#E8EAF6,stroke:#3F51B5,stroke-width:1px;
    classDef device fill:#F1F8E9,stroke:#558B2F,stroke-width:1px;
    classDef service fill:#FFF3E0,stroke:#E65100,stroke-width:1px;
    classDef db fill:#F3E5F5,stroke:#6A1B9A,stroke-width:1px;
    classDef auth fill:#E1BEE7,stroke:#4A148C,stroke-width:1px;

    User["Kasutaja<br/>Brauser<br/>(avalik IP)"]:::user
    Internet((Internet)):::network
    Router["Kodune ruuter (Cudy)<br/>(NAT, väline IP)<br/>Port forward -> ESP32:8080"]:::network
    ESP32["ESP32 seade<br/>(VPN klient: 10.8.0.2)<br/>Web server :8080"]:::device

    subgraph DO["Digital Ocean droplet<br/>(VPN server + Keskserver + MQTT broker + InfluxDB)<br/>Public IP: 143.198.250.135<br/>VPN GW: 10.8.0.1"]
        VPN["OpenVPN server"]:::auth
        Caddy["Caddy reverse proxy<br/>TLS :443<br/>hello.narbot.ee<br/>dashboard.narbot.ee<br/>esp.narbot.ee"]:::service
        Flask["Flask Keskserver<br/>:8000 (reverse proxy kaudu)"]:::service
        MQTT["Mosquitto MQTT broker<br/>TCP :1883"]:::service
        Influx["InfluxDB<br/>HTTP API :8086"]:::db
    end
    class DO cloud;

    User -->|HTTPS :443| Internet
    Internet --> Router
    Router --> ESP32
    Internet --> DO

    ESP32 -->|MQTT publish :1883 VPN<br/>Staatuse sõnumid| MQTT
    MQTT -->|Forward| Flask
    Flask -->|Write| Influx
    User -->|Seadmete nimekiri<br/>/ UI| Caddy
    Caddy -->|reverse_proxy 127.0.0.1:8000| Flask
    Caddy -->|reverse_proxy 10.8.0.2:8080| ESP32

    %% Auth layers
    ESP32 -->|VPN autentimine<br/>OpenVPN| VPN
```

User -->|Keskserveri login
Flask username/password| Flask
ESP32 -->|MQTT auth
hetkel puudub| MQTT

Software stack diagramm

ESP32 stack

flowchart TB

```
classDef layer fill:#F1F8E9,stroke:#558B2F,stroke-width:1px;
classDef mid fill:#E3F2FD,stroke:#1565C0,stroke-width:1px;
classDef hw fill:#ECEFF1,stroke:#37474F,stroke-width:1px;
```

```
ESPLogic["Loogika (Arduino)<br/>- Valve control (GPIO)<br/>- Pressure reading (ADC)<br/>- MQTT publishing"]:::layer
ESPLibs["Library kihid<br/>- LittleFS (HTML template'id)<br/>- PubSubClient (MQTT klient)<br/>- NTPClient (aja sünkronisatsioon)"]:::layer
ESPFW["Arduino/ESP-IDF framework"]:::mid
ESPHW["ESP32 Hardware<br/>(WiFi, ADC, GPIO, Flash, NVS)"]:::hw
```

ESPLogic --> ESPLibs --> ESPFW --> ESPHW

```
ESPLogic -->|MQTT publish :1883| MQTTBroker["Mosquitto MQTT Broker<br/>D0 droplet"]:::mid
```

Keskserveri stack

flowchart TB

```
classDef ui fill:#FFF3E0,stroke:#E65100,stroke-width:1px;
classDef service fill:#E8EAF6,stroke:#3F51B5,stroke-width:1px;
classDef db fill:#F3E5F5,stroke:#6A1B9A,stroke-width:1px;
classDef mqtt fill:#E0F7FA,stroke:#00796B,stroke-width:1px;
```

```
Browser["Kasutaja brauser<br/>(HTML, CSS, Chart.js)"]:::ui
Caddy["Caddy reverse proxy<br/>TLS :443"]:::service
Flask["Flask veebirakendus<br/>Routes:<br/>- /<br/>- /device/&lt;id&gt;/history<br/>- /export"]:::service
Jinja["Jinja2 template rendering"]:::service
InfluxClient["InfluxDB client library<br/>(päringud, lugemine)"]:::service
Influx["InfluxDB :8086<br/>Bucket: syringe_bucket<br/>Measurement: device_status"]:::db
Listener["MQTT Listener<br/>(Python script/thread)"]:::mqtt
Broker["Mosquitto MQTT Broker :1883"]:::mqtt
ESP32["ESP32 seadmed<br/>(PubSubClient)"]:::mqtt
```

Browser -->|HTTPS :443| Caddy
Caddy -->|reverse_proxy 127.0.0.1:8000| Flask
Caddy -->|reverse_proxy 10.8.0.2:8080| ESP32
Flask --> Jinja --> InfluxClient --> Influx
Listener --> Influx
Broker --> Listener

```
ESP32 -->|MQTT publish| Broker
class Caddy service;
```

Suhtlus (sequence diagrammid)

Stsenaarium 1: ESP32 saadab staatust

```
sequenceDiagram
    participant ESP32 as ESP32
    participant MQTT as MQTT Broker
    participant Listener as MQTT Listener
    participant Influx as InfluxDB

    ESP32->>ESP32: mõõtmine → buffer
    ESP32->>MQTT: publish buffer[0] (VPN)
    MQTT-->>ESP32: OK
    ESP32->>ESP32: eemalda buffer[0]
    MQTT->>Listener: forward
    Listener->>Influx: write

    ESP32->>ESP32: mõõtmine → buffer
    ESP32->>MQTT: publish buffer[0]
    MQTT--x ESP32: WiFi katkeb, FAIL

    ESP32->>ESP32: mõõtmine → buffer
    ESP32->>ESP32: mõõtmine → buffer

    Note over ESP32: WiFi taastub
    ESP32->>MQTT: publish buffer[0]
    MQTT-->>ESP32: OK
    ESP32->>ESP32: eemalda buffer[0]
    ESP32->>MQTT: publish buffer[1]
    MQTT-->>ESP32: OK
    ESP32->>ESP32: eemalda buffer[1]
```

Stsenaarium 2: Kasutaja vaatab seadmete nimekirja

```
sequenceDiagram
    participant Browser as Browser
    participant Flask as Flask Server
    participant Influx as InfluxDB

    Browser->>Caddy: GET /
    Caddy->>Flask: reverse_proxy 127.0.0.1:8000
    Flask->>Influx: query last status
    Influx-->>Flask: result
    Flask-->>Browser: HTML + data (Jinja2)
```

Stsenaarium 3: Kasutaja juhib ESP32 klappi

```
sequenceDiagram
    participant Button as Nupp
```

```

participant ESP32 as ESP32 Loogika
participant Caddy as Caddy
participant MQTT as MQTT Broker
participant Listener as MQTT Listener
participant Influx as InfluxDB

Button->>ESP32: press (GPIO interrupt)
ESP32->>ESP32: toggle valve state (GPIO)
ESP32->>MQTT: publish topic sensors/{uid}/
status\n{"valve_state":"open","pressure_30ms_ago":2.1,"pressure_now":4.8}
MQTT->>Listener: forward
Listener->>Influx: write Point(...)

```

Tehnoloogiate legend

- OpenVPN: VPN protokoll (UDP/TCP)
- MQTT: Message broker protokoll (TCP, port 1883)
- MQTT autentimine: hetkel puudub
- Mosquitto: MQTT broker tarkvara
- InfluxDB: Time-series andmebaas (HTTP API, port 8086)
- Caddy: Reverse proxy + TLS terminatsioon (HTTPS :443)
- Flask: Python web framework
- Jinja2: Template rendering (Flask sees)
- NTP: Network Time Protocol (aja sünkronisatsioon)
- LittleFS: ESP32 failisüsteem
- NVS: ESP32 non-volatile storage (WiFi kredentsiaalid)
- PubSubClient: Arduino MQTT kliendi library
- Chart.js: JavaScript graafikute teek

Miks need diagrammid on olulised

Need diagrammid dokumenteerivad süsteemi arhitektuuri viisil, mida on võimalik:

- Näidata, et sa mõistad kogu süsteemi terviklikult
- Kasutada ise tulevikus kui pead süsteemi uuesti üles seadma
- Jagada teiste tudengitega, et nad saaksid sinu lahendusest õppida
- Kasutada alusena kui tahad süsteemi laiendada (nt lisada uusi sensoreid)

Diagrammid peaksid olema piisavalt detailsed, et keegi teine saaks nende põhjal süsteemi üles ehitada, kuid mitte nii detailsed, et muutuvad loetamatuks. Kasuta värve, et eristada erinevaid komponente ja andmevooge.